

EE-482: Network Security and Cryptography

LECTURES NO 8

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Organization of Lecture

1. Advanced Encryption Standard (AES)
2. Blowfish Algorithm
3. IDEA Algorithm

1. Advanced Encryption Standard (AES) [1/3]

↳ Symmetric key Block cipher (same key is used for encryption and decryption).

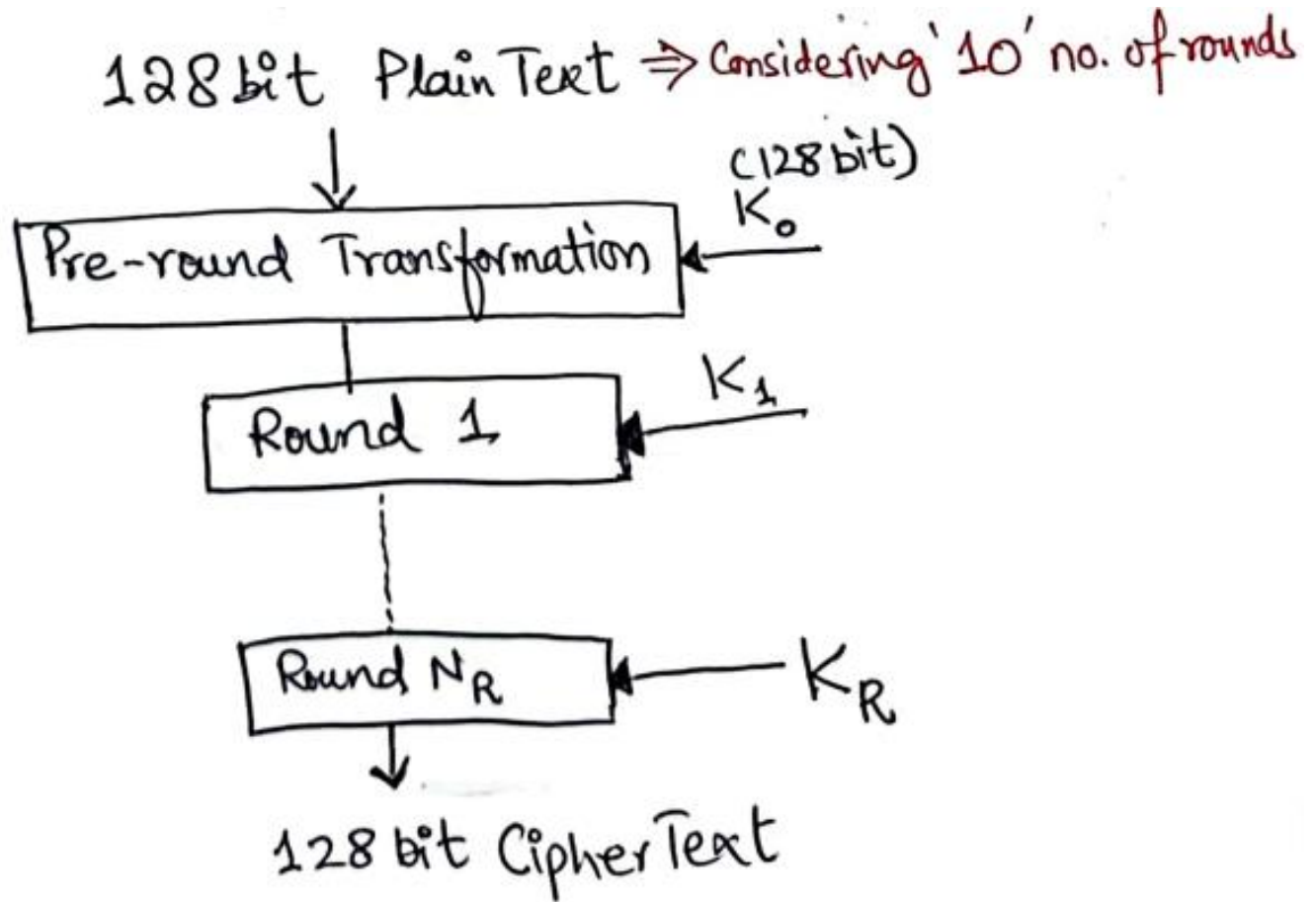
↳ 128-bit data & 128-bit key.

↳ Stronger and faster than DES.

<u>No. of Rounds</u>	<u>Key used</u>
10	128
12	192
14	256

↳ 128-bit data is divided into blocks of 16 Bytes.

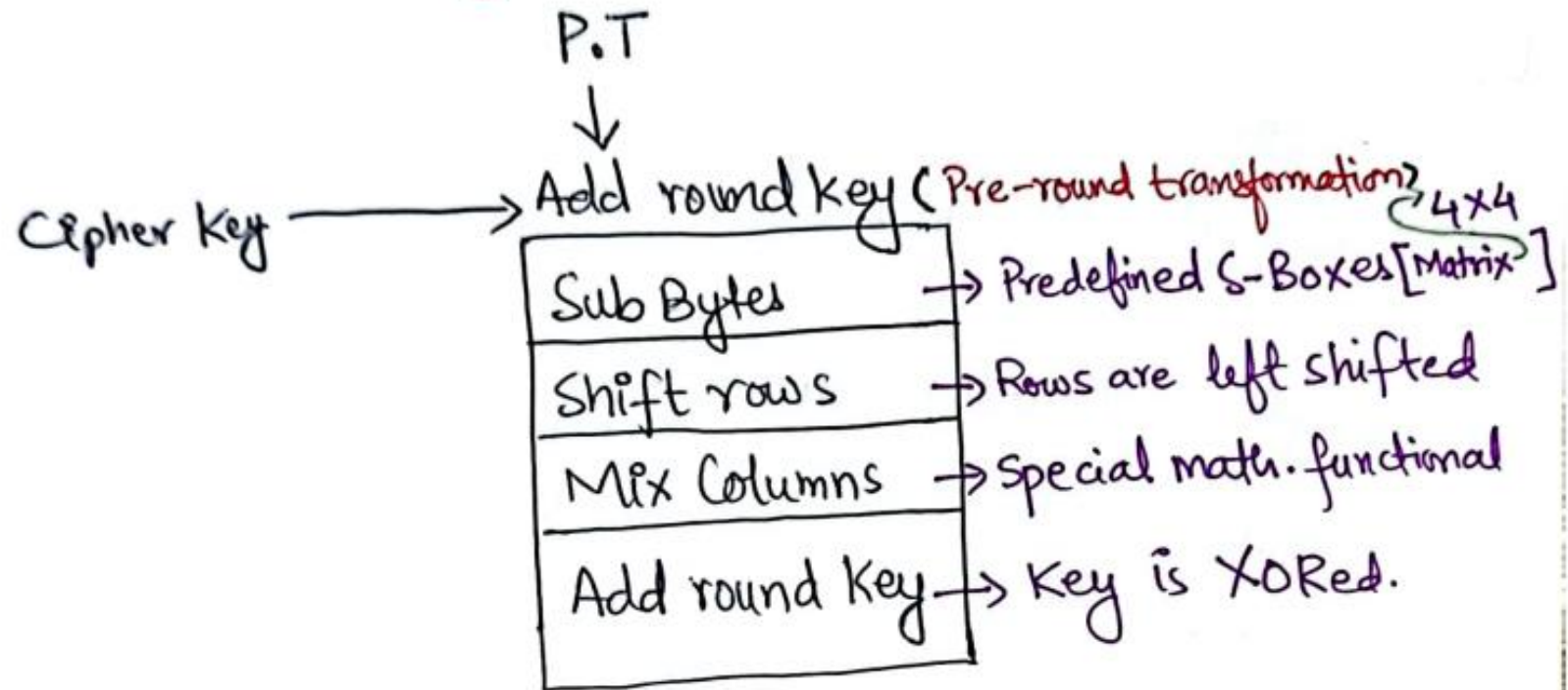
1. Advanced Encryption Standard (AES) [2/3]



1. Advanced Encryption Standard (AES) [3/3]

↳ Complete encryption is performed in round process.

Encryption Process:- (128 bit Key/10 no. of Rounds)



Add round Key: XOR between P.T and Cipher Key

2. Blowfish Algorithm [1/6]

↳ Developed by Bruce Schneier.

↳ Symmetric Key Algorithm.

↳ Can be seen as an alternative to DES & IDEA Algorithms.

↳ It is a block Cipher Algorithm.

↳ Some Properties:-

- (i) Fast
- (ii) Compact → execute in less memory.
- (iii) Simple → Basic operations → XOR, Addition.
- (iv) Secure → Variable length key.

↳ Blowfish algorithm encrypts 64-bit block with variable length key.

↳ Key size can be from 32 bits to **448 bits**

2. Blowfish Algorithm [2/6]

Parts of Blowfish Algorithm:-

- i) Subkey Generation
- ii) Data Encryption.

Subkey Generation:-

→ Keys should be ready before encryption/decryption process.

- i) $K_1, K_2, \dots, K_n$ [$1 \leq n \leq 14$] ; Placed in an array & divided into 14 words. Each word has a length of 32 bits i.e.,

$$32 \times 14 = 448 \text{ bits}$$

- ii) Initialize a P-Array

$P_1, P_2, \dots, P_{18}$ [32-bit length → each word].
↳ (total 18 words)

2. Blowfish Algorithm [3/6]

iii) Initialize 4-S boxes [Each S-box contains 256, 32-bit entries]
where

$S_1 \rightarrow S_0, \dots, S_{255}$ (S_1 will have 256 bits)
 $S_2 \rightarrow \dots$ (S_2 " " " ")
 $S_3 \rightarrow \dots$ (S_3 " " " ")
 $S_4 \rightarrow \dots$ (S_4 " " " ")

iv) P-Array, S-Boxes are initialized using hexadecimal representation.

v) $P_1 = P_1 \text{ XOR } K_1$

$P_2 = P_2 \text{ XOR } K_2$

$P_{14} = P_{14} \text{ XOR } K_{14}$ $\rightarrow$ Max. size is K_{14}

$P_{15} = P_{15} \text{ XOR } K_1$

$P_{16} = P_{16} \text{ XOR } K_2$

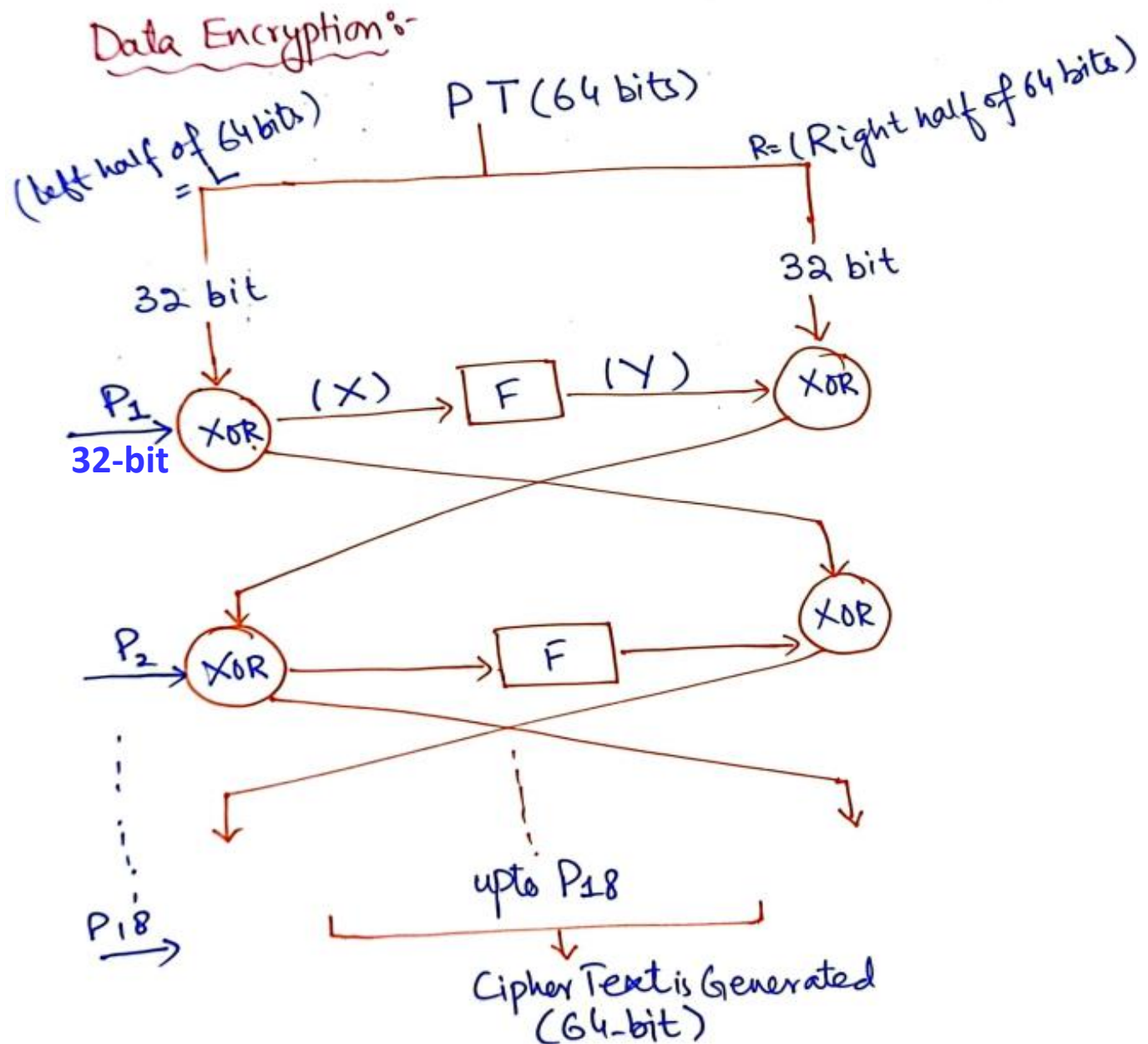
$P_{18} = P_{18} \text{ XOR } K_4$

2. Blowfish Algorithm [4/6]

vi) Take 64-bit Plain Text & initially all 64-bits be Zero, i.e.,
(0, 0)

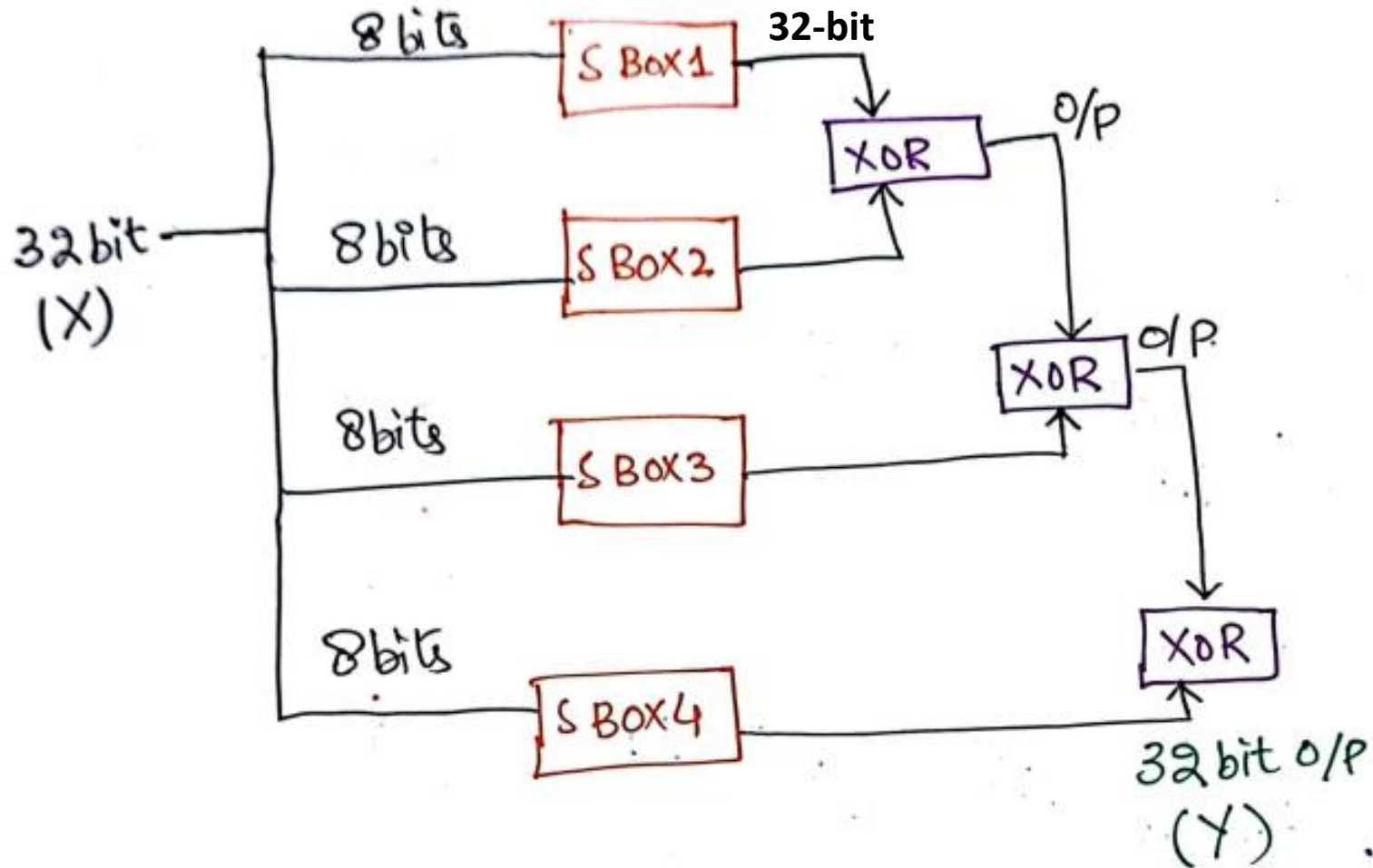
↓
Blowfish Encry. → Generates Subkeys:

2. Blowfish Algorithm [5/6]



2. Blowfish Algorithm [6/6]

What happens in side function block 'F'?

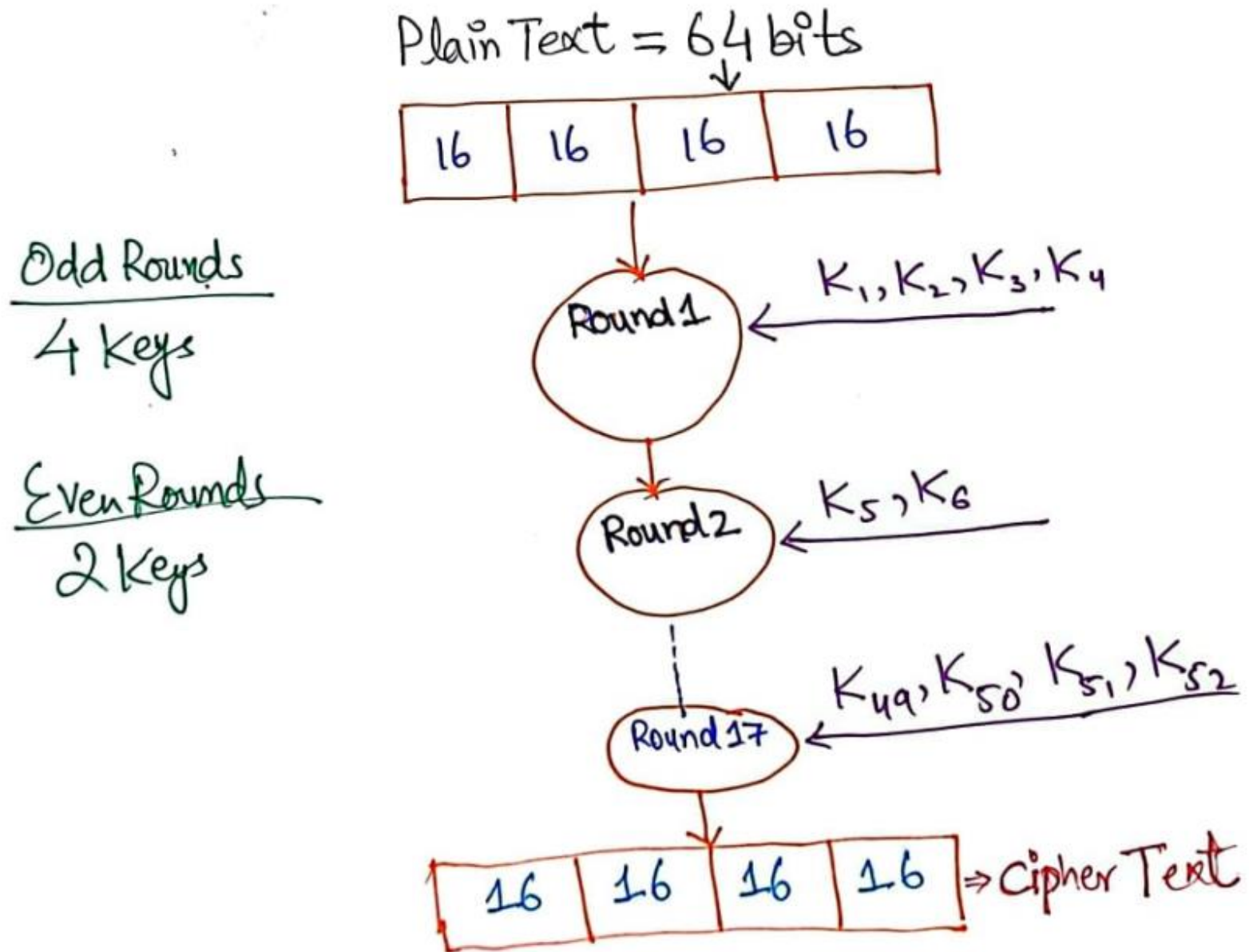


3. IDEA Algorithm [1/6]

3) IDEA Algorithm

- ↳ International Data Encryption Algorithm (IDEA)
- ↳ It is a block cipher Algorithm.
- ↳ Symmetric Key Algorithm.
- ↳ It is a Feistel cipher (It is not a specific scheme of block cipher. It is a design model from which many different block ciphers are derived).
- ↳ 64 bit Plain text $\Rightarrow$ 16, 16, 16, 16.
- ↳ Key size = 128 bits $\Rightarrow$ divided into '52' sub keys.
- ↳ There are total '17' rounds.

3. IDEA Algorithm [2/6]



3. IDEA Algorithm [3/6]

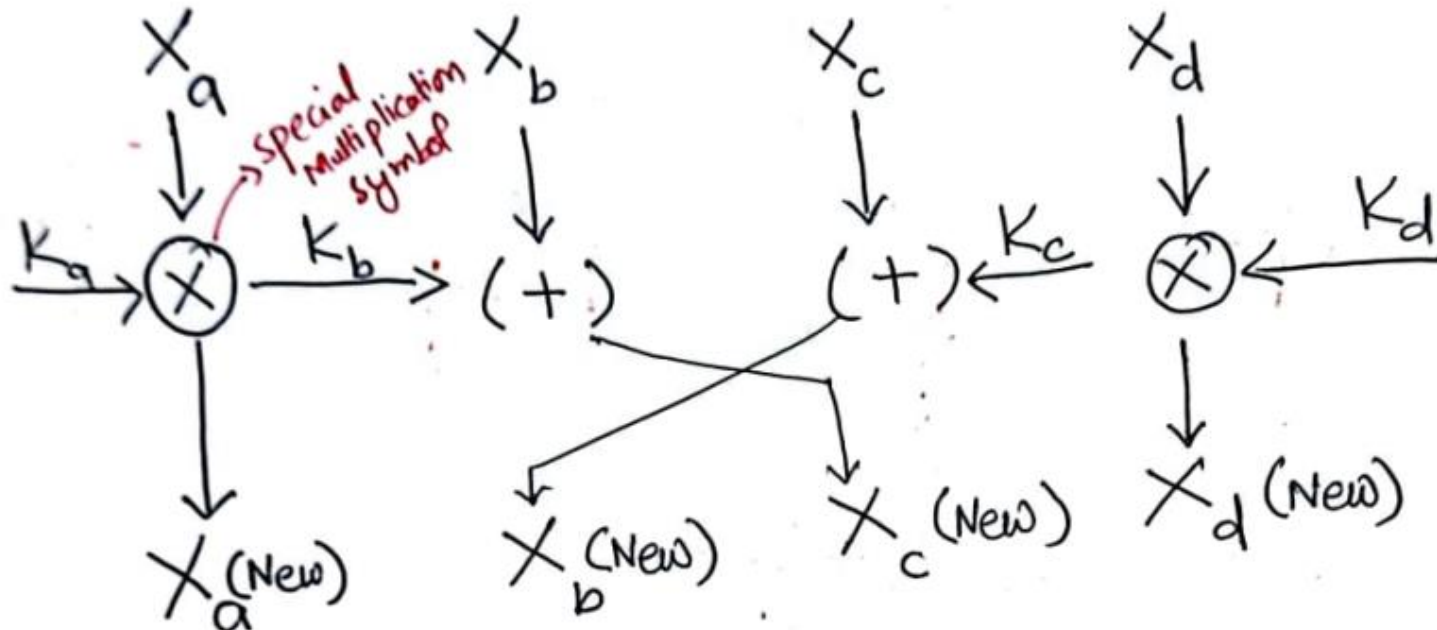
↳ For even number of rounds we will have one procedure and for odd number of rounds we have another procedure.

Rounds

* Odd Rounds (1, 3, 5, 7, ..., 17)

↳ Number of input ^{parts} = 4 $\Rightarrow \{X_a, X_b, X_c, X_d\}$

↳ Number of keys = 4 $\Rightarrow \{K_a, K_b, K_c, K_d\}$



3. IDEA Algorithm [4/6]

* Even Rounds (2, 4, 6, 8, 16)

- ↳ Number of Input parts = 4 $\Rightarrow \{X_a, X_b, X_c, X_d\}$
- ↳ " " Keys = 2 $= \{K_e, K_f\}$

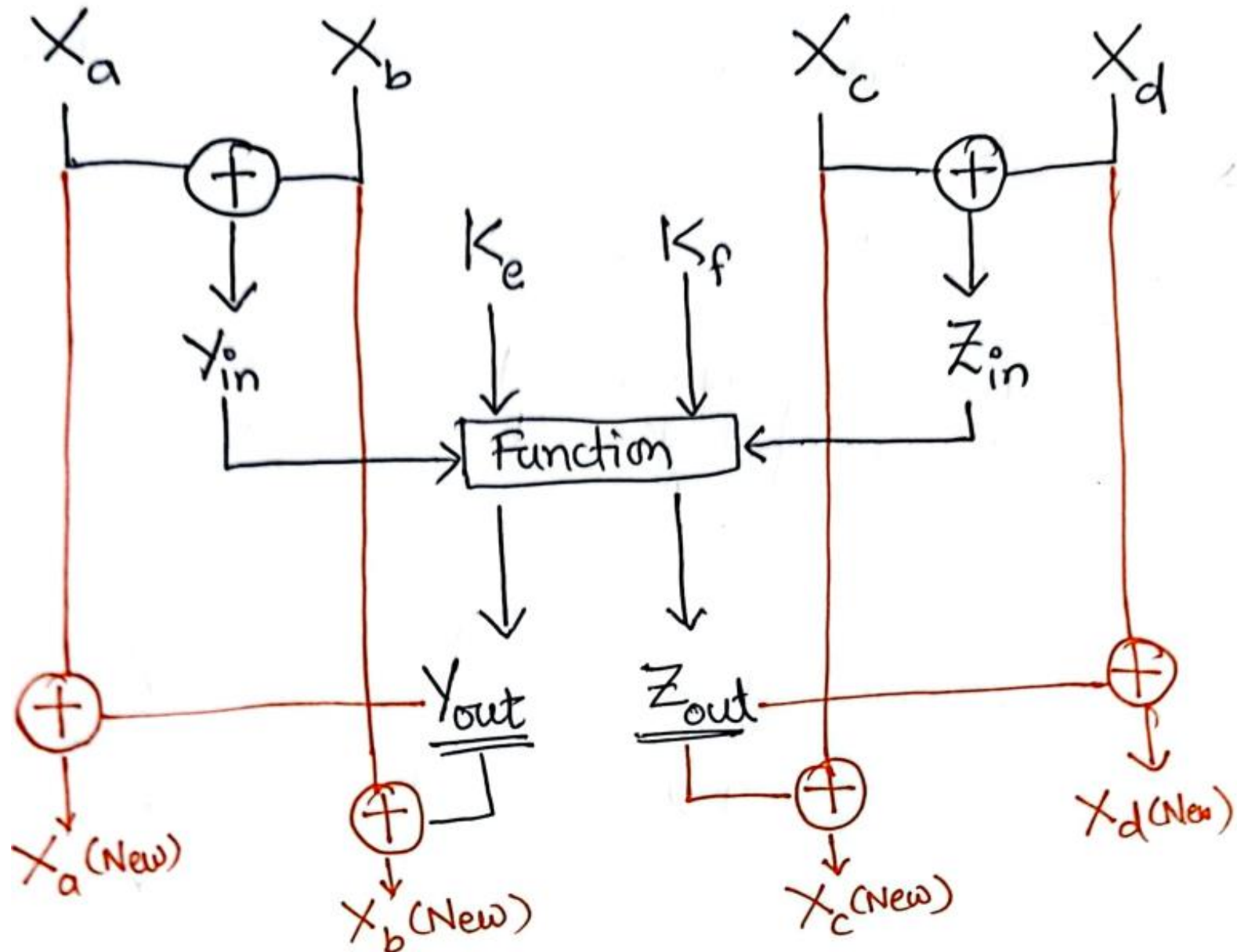
$$\begin{array}{cc} X_a \oplus X_b & X_c \oplus X_d \\ \underbrace{\quad \quad}_{\text{1st parameter}} & \underbrace{\quad \quad}_{\text{2nd parameter}} \\ K_e, & K_f \end{array}$$

↳ 1st parameter = $Y_{in}^0 = X_a \oplus X_b$

↳ 2nd parameter = $Z_{in}^0 = X_c \oplus X_d$

Now we have 2 Inputs (2 parameters) + 2 Keys, we can perform rounds easily

3. IDEA Algorithm [5/6]



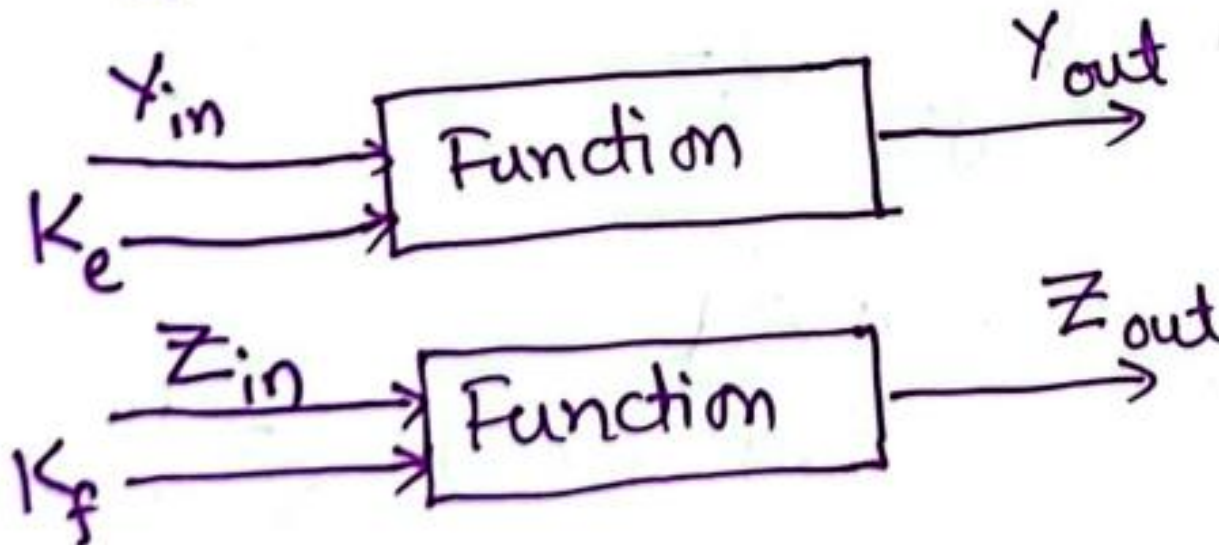
3. IDEA Algorithm [6/6]

$$* X_a(\text{New}) = X_a \oplus Y_{\text{out}}$$

$$* X_b(\text{New}) = X_b \oplus Y_{\text{out}}$$

$$* X_c(\text{New}) = X_c \oplus Z_{\text{out}}$$

$$* X_d(\text{New}) = X_d \oplus Z_{\text{out}}$$



References: Textbooks

1. *“Cryptography and Network Security”, by Behrouz A. Forouzan, Debdeep Mukhopadhyay, 2nd Edition, McGraw Hill Education (India) Private Limited.*
2. *“Cryptography and Network Security Principles and Practice”, by William Stallings, 6th Edition, Pearson.*
3. *“Computer Security Principles and Practice”, by William Stallings, Lawrie Brown, 4th Edition, Pearson.*